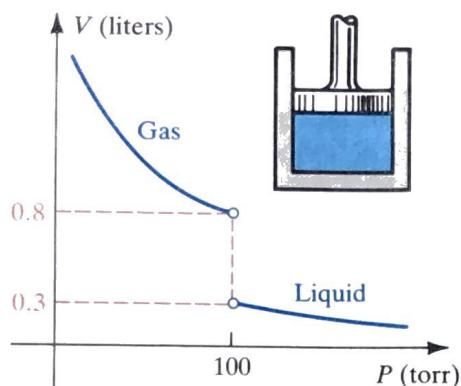


The following application involves one-sided limits.

FIGURE 2.9



EXAMPLE 8 A gas (such as water vapor or oxygen) is held at a constant temperature in the piston shown in Figure 2.9. As the gas is compressed, the volume V decreases until a certain critical pressure is reached. Beyond this pressure, the gas will assume liquid form. Use the graph in Figure 2.9 to find and interpret

(a) $\lim_{P \rightarrow 100^-} V$ (b) $\lim_{P \rightarrow 100^+} V$ (c) $\lim_{P \rightarrow 100} V$

SOLUTION

(a) We see from Figure 2.9 that when the pressure P (in torr) is low, the substance is a gas and the volume V (in liters) is large. (The definition of the unit of pressure, the *torr*, may be found in textbooks on physics.) If P approaches 100 through values less than 100, V decreases and approaches 0.8; that is,

$$\lim_{P \rightarrow 100^-} V = 0.8.$$

The limit 0.8 represents the volume at which the substance begins to change from a gas to a liquid.

(b) If $P > 100$, the substance is a liquid. If P approaches 100 through values greater than 100, the volume V increases very slowly (since liquids are nearly incompressible), and

$$\lim_{P \rightarrow 100^+} V = 0.3.$$

The limit 0.3 represents the volume at which the substance begins to change from a liquid to a gas.

(c) $\lim_{P \rightarrow 100} V$ does not exist since the left-hand and right-hand limits in (a) and (b) are not equal. (At $P = 100$ the gas and liquid forms exist together in equilibrium, and the substance cannot be classified as either a gas or a liquid.)

EXERCISES 2.1

Exer. 1–10: Find the limit.

1 $\lim_{x \rightarrow -2} (3x - 1)$

2 $\lim_{x \rightarrow 3} (x^2 + 2)$

3 $\lim_{x \rightarrow 4} x$

4 $\lim_{x \rightarrow -3} (-x)$

5 $\lim_{x \rightarrow 100} 7$

6 $\lim_{x \rightarrow 7} 100$

7 $\lim_{x \rightarrow -1} \pi$

8 $\lim_{x \rightarrow \pi} (-1)$

9 $\lim_{x \rightarrow -1} \frac{x + 4}{2x + 1}$

10 $\lim_{x \rightarrow 5} \frac{x + 2}{x - 4}$

Exer. 11–24: Use an algebraic simplification to help find the limit, if it exists.

11 $\lim_{x \rightarrow -3} \frac{(x + 3)(x - 4)}{(x + 3)(x + 1)}$

12 $\lim_{x \rightarrow -1} \frac{(x + 1)(x^2 + 3)}{x + 1}$

13 $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

14 $\lim_{x \rightarrow 3} \frac{2x^3 - 6x^2 + x - 3}{x - 3}$

15 $\lim_{r \rightarrow 1} \frac{r^2 - r}{2r^2 + 5r - 7}$

16 $\lim_{r \rightarrow -3} \frac{r^2 + 2r - 3}{r^2 + 7r + 12}$

17 $\lim_{k \rightarrow 4} \frac{k^2 - 16}{\sqrt{k} - 2}$

18 $\lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25}$

$$19 \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$21 \lim_{h \rightarrow -2} \frac{h^3 + 8}{h + 2}$$

$$23 \lim_{z \rightarrow -2} \frac{z - 4}{z^2 - 2z - 8}$$

$$20 \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

$$22 \lim_{h \rightarrow 2} \frac{h^3 - 8}{h^2 - 4}$$

$$24 \lim_{z \rightarrow 5} \frac{z - 5}{z^2 - 10z + 25}$$

Exer. 25–30: Find each limit, if it exists:

(a) $\lim_{x \rightarrow a^-} f(x)$ (b) $\lim_{x \rightarrow a^+} f(x)$ (c) $\lim_{x \rightarrow a} f(x)$

$$25 f(x) = \frac{|x - 4|}{x - 4}; \quad a = 4$$

$$26 f(x) = \frac{x + 5}{|x + 5|}; \quad a = -5$$

$$27 f(x) = \sqrt{x + 6} + x; \quad a = -6$$

$$28 f(x) = \sqrt{5 - 2x} - x^2; \quad a = \frac{5}{2}$$

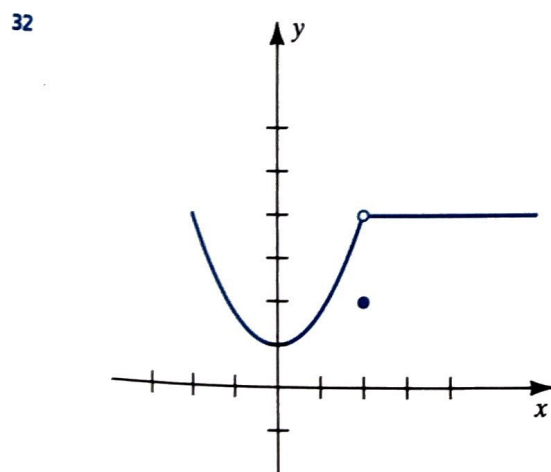
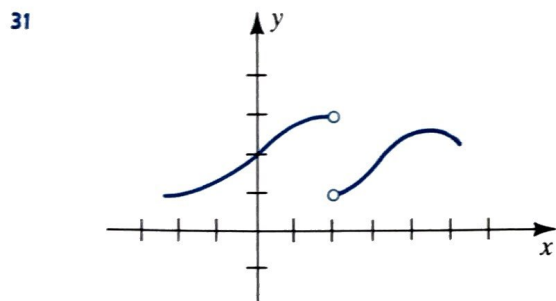
$$29 f(x) = \frac{1}{x^3}; \quad a = 0$$

$$30 f(x) = \frac{1}{x - 8}; \quad a = 8$$

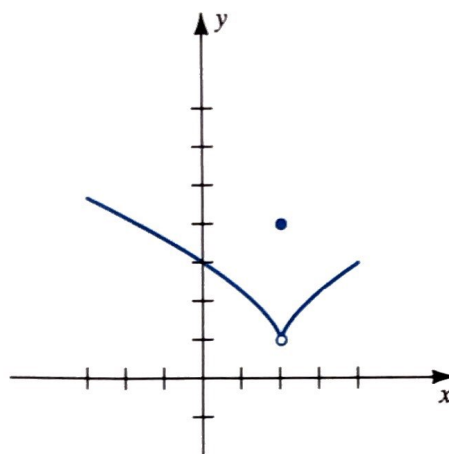
Exer. 31–40: Refer to the graph to find each limit, if it exists:

(a) $\lim_{x \rightarrow 2^-} f(x)$ (b) $\lim_{x \rightarrow 2^+} f(x)$ (c) $\lim_{x \rightarrow 2} f(x)$

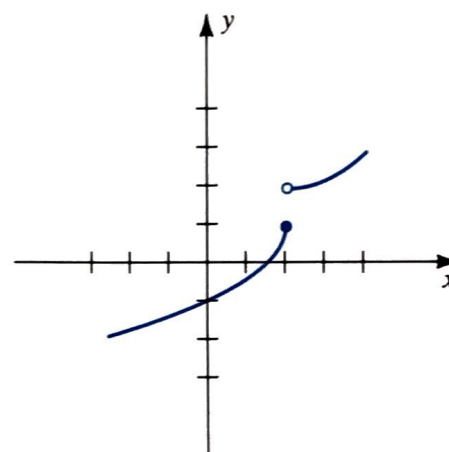
(d) $\lim_{x \rightarrow 0^-} f(x)$ (e) $\lim_{x \rightarrow 0^+} f(x)$ (f) $\lim_{x \rightarrow 0} f(x)$



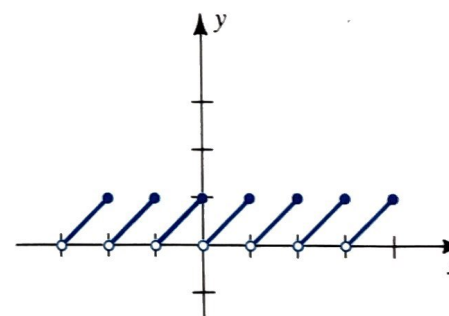
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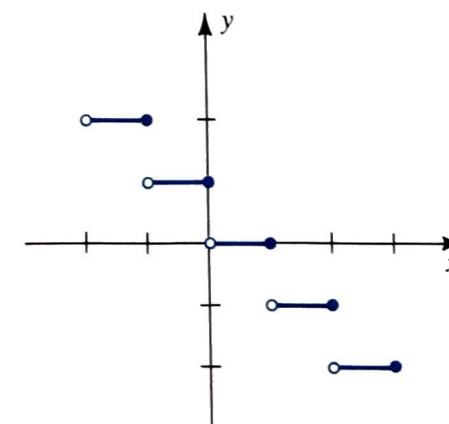
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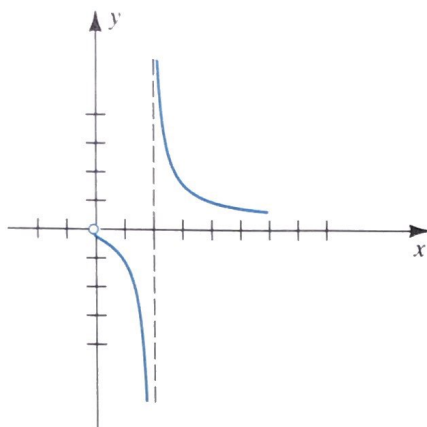
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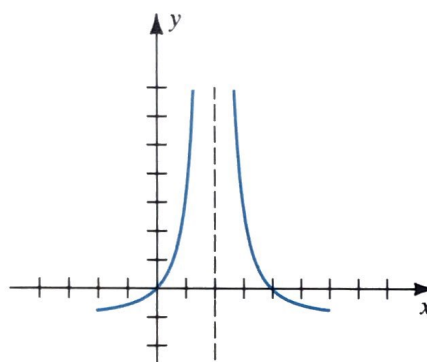
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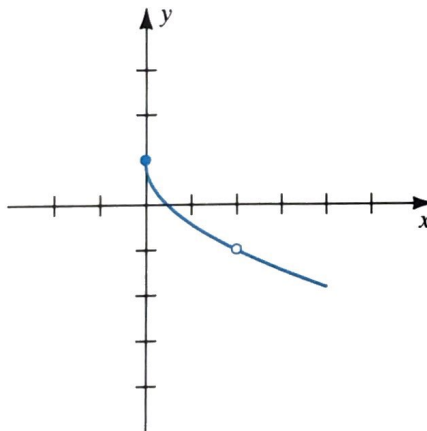
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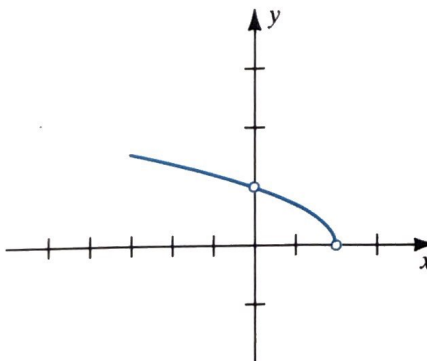
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Exer. 41–46: Sketch the graph of f and find each limit, if it exists:

(a) $\lim_{x \rightarrow 1^-} f(x)$ (b) $\lim_{x \rightarrow 1^+} f(x)$ (c) $\lim_{x \rightarrow 1} f(x)$

41 $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 1 \\ 4 - x & \text{if } x \geq 1 \end{cases}$

42 $f(x) = \begin{cases} x^3 & \text{if } x \leq 1 \\ 3 - x & \text{if } x > 1 \end{cases}$

43 $f(x) = \begin{cases} 3x - 1 & \text{if } x \leq 1 \\ 3 - x & \text{if } x > 1 \end{cases}$

44 $f(x) = \begin{cases} |x - 1| & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$

45 $f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ 1 & \text{if } x = 1 \\ x + 1 & \text{if } x > 1 \end{cases}$

46 $f(x) = \begin{cases} -x^2 & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ x - 2 & \text{if } x > 1 \end{cases}$

47 A country taxes the first \$20,000 of an individual's income at a rate of 15%, and all income over \$20,000 is taxed at 20%.

(a) Find a piecewise-defined function T for the total tax on an income of x dollars.

(b) Find

$$\lim_{x \rightarrow 20,000^-} T(x) \quad \text{and} \quad \lim_{x \rightarrow 20,000^+} T(x).$$

48 A telephone company charges 25 cents for the first minute of a long-distance call and 15 cents for each additional minute.

(a) Find a piecewise-defined function C for the total cost of a long-distance call of x minutes.

(b) If n is an integer greater than 1, find

$$\lim_{x \rightarrow n^-} C(x) \quad \text{and} \quad \lim_{x \rightarrow n^+} C(x).$$

49 Shown in the figure on the next page is a graph of the g -forces experienced by astronauts during the takeoff of a spacecraft with two rocket boosters. (A force of $2g$'s is twice that of gravity, $3g$'s is three times that of gravity, etc.) If $F(t)$ denotes the g -force t minutes into the flight, find and interpret

(a) $\lim_{t \rightarrow 0^+} F(t)$

(b) $\lim_{t \rightarrow 3.5^-} F(t)$ and $\lim_{t \rightarrow 3.5^+} F(t)$

(c) $\lim_{t \rightarrow 5^-} F(t)$ and $\lim_{t \rightarrow 5^+} F(t)$